EN3160: Image Processing and Machine Vision

Assignment 2: Fitting and Alignment

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October 18, 2026

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GitHub Repository: github.com/RumethRanhinda/EN3160-Vision-UoM

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1. Blob Detection using Laplacian of Gaussians

In this task, we aim to detect sunflowers by employing normalized Laplacian of Gaussians (LoG). Due to flowers with varying sizes, a range of σ values must be used. This algorithm will look for dark seed heads which is characteristic for a sunflower.

1.1 Scale (σ) and Radius Selection

The sunflower field image is loaded with `cv.IMREAD_REDUCED_COLOR_4`, reducing dimensions to 360×360 px (1/4 th). In this scale:

- **Scale Range ($\sigma \in [1.20, 8.00]$):** Distant sunflower blooms near the horizon have dark centers spanning $r_{\min} \approx 1.7$ px. From the condition $r = \sqrt{2}\sigma$, this gives $\sigma_{\min} = 1.7/\sqrt{2} \approx 1.20$. The largest foreground sunflower seed heads span $r_{\max} \approx 10$–11.3 px, setting $\sigma_{\max} = 11.31/\sqrt{2} \approx 8.00$.
- **Scale Progression:** A geometric progression of $S = 24$ levels ($\sigma_i = \sigma_{\min} k^i$ with $k = 1.086$) provides coverage from $r = 1.70$ px to 11.31 px.
- **Extrema Detection:** Scale normalized LoG candidate peaks are extracted via 3D Non-Maximum Suppression ($3 \times 3 \times 3$ window, $\tau = 0.05$). The RoI ignored the image above the horizon as to avoid large trees in the distance to be considered as a blob.

[21]:
```
import cv2
import numpy as np
import scipy.ndimage as ndi
import matplotlib.pyplot as plt

# Load image
img_bgr = cv2.imread('src/the_berry_farms_sunflower_field.jpeg', cv2.IMREAD_REDUCED_COLOR_4)
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY).astype(np.float64) / 255.0
hsv = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2HSV)
h, w = gray.shape

# Yellow petal mask for colour confirmation
yellow_petals = (hsv[:, :, 0] >= 14) & (hsv[:, :, 0] <= 38) & (hsv[:, :, 1] >= 40) & (hsv[:, :,␣
↪2] >= 40)
```

```
# Define scale space LoG
sigmas = np.geomspace(1.20, 8.00, 24)
scale_space = np.zeros((len(sigmas), h, w), dtype=np.float64)
for i, s in enumerate(sigmas):
    scale_space[i] = (s ** 2) * ndi.gaussian_laplace(gray, sigma=s)

# 3D Non Maximum Suppression across (scale, y, x)
footprint = np.ones((3, 3, 3), dtype=bool)
max_vol = ndi.maximum_filter(scale_space, footprint=footprint, mode='constant', cval=0.0)

# Exclude above horizon trees
peaks = (scale_space == max_vol) & (scale_space > 0.05)
peaks[0, :, :] = False
peaks[-1, :, :] = False
peaks[:, :182, :] = False

# Filter candidate peaks to confirm sunflowers
blobs = []
for s_idx, y, x in np.argwhere(peaks):
    if 0 < x < (w - 1) and 0 < y < (h - 1):
        s = sigmas[s_idx]
        blob_r = np.sqrt(2) * s
        score = scale_space[s_idx, y, x]

        # Reject green leaf foliage centers
        c_h, c_s, c_v = hsv[y, x]
        if 35 < c_h < 85 and c_v > 65:
            continue

        # Check surrounding annulus for yellow petals
        r_in = max(1.0, 0.7 * blob_r)
        r_out = min(25.0, 2.2 * blob_r)
        x_min = max(0, int(x - r_out)); x_max = min(w, int(x + r_out + 1))
        y_min = max(0, int(y - r_out)); y_max = min(h, int(y + r_out + 1))

        Y_sub, X_sub = np.ogrid[y_min:y_max, x_min:x_max]
        d2 = (X_sub - x) ** 2 + (Y_sub - y) ** 2
        annulus = (d2 >= r_in ** 2) & (d2 <= r_out ** 2)

        petal_cnt = np.sum(yellow_petals[y_min:y_max, x_min:x_max] & annulus)
        ann_area = np.sum(annulus)
        petal_frac = petal_cnt / max(1, ann_area)

        # Distant small blooms need minimal petal support
        if blob_r <= 3.5:
            if petal_cnt >= 1 or petal_frac >= 0.03:
                blobs.append({'x': float(x), 'y': float(y), 'r': float(blob_r), 'sigma':␣
↪float(s), 'score': float(score)})
        else:
            if petal_cnt >= 8 and petal_frac >= 0.20 and score >= 0.20:
                blobs.append({'x': float(x), 'y': float(y), 'r': float(blob_r), 'sigma':␣
↪float(s), 'score': float(score)})

# Extract top 5 largest sunflowers
sorted_blobs = sorted(blobs, key=lambda b: (b['r'], b['score']), reverse=True)
largest_circles = []
for b in sorted_blobs:
```

```
    too_close = any(np.hypot(b['x'] - k['x'], b['y'] - k['y']) < k['r'] * 0.7 for k in␣
 ↪largest_circles)
    if not too_close:
        largest_circles.append(b)
    if len(largest_circles) == 5:
        break

print(f"Total detected blobs: {len(blobs)}")
print(f"Sigma range used: [{sigmas[0]:.2f}, {sigmas[-1]:.2f}] across {len(sigmas)} levels (radii:
 ↪ [{np.sqrt(2)*sigmas[0]:.2f}, {np.sqrt(2)*sigmas[-1]:.2f}] px)")
print("\nParameters of the 5 Largest Circles:")
print(f"{'Rank':<6}{'Center (x, y)':<18}{'Radius r (px)':<16}{'Scale sigma':<14}{'LoG Response':
 ↪<12}")
print("-" * 66)
for i, b in enumerate(largest_circles):
    print(f"{i+1:<6}({b['x']:.1f}, {b['y']:.1f}){'':<6}{b['r']:<16.2f}{b['sigma']:<14.
 ↪2f}{b['score']:<12.4f}")

# Display
fig, ax = plt.subplots(figsize=(6.5, 6.5))
ax.imshow(img_rgb)
for b in blobs:
    ax.add_patch(plt.Circle((b['x'], b['y']), b['r'], color='yellow', fill=False, linewidth=0.8,␣
 ↪alpha=0.75))

for i, b in enumerate(largest_circles):
    ax.add_patch(plt.Circle((b['x'], b['y']), b['r'], color='red', fill=False, linewidth=2.0))
    ax.text(b['x'], b['y'], f"#{i+1}", color='white', fontsize=8, fontweight='bold',␣
 ↪ha='center', va='center',
            bbox=dict(boxstyle='circle,pad=0.2', facecolor='red', alpha=0.9, edgecolor='none'))

ax.set_title(f"Detected Sunflowers (Top 5 Largest in Red)", fontsize=10.5, pad=6)
ax.axis('off')
plt.tight_layout()
plt.show()
```

```
Total detected blobs: 361
Sigma range used: [1.20, 8.00] across 24 levels (radii: [1.70, 11.31] px)

Parameters of the 5 Largest Circles:
Rank  Center (x, y)     Radius r (px)   Scale sigma   LoG Response
------------------------------------------------------------------
1     (282.0, 338.0)      10.42           7.37          0.3489
2     (106.0, 255.0)      8.13            5.75          0.3632
3     (178.0, 261.0)      8.13            5.75          0.3515
4     (179.0, 327.0)      8.13            5.75          0.3440
5     (189.0, 240.0)      6.35            4.49          0.3648
```



Observations: A total of 408 blobs were detected across the field. All top 5 largest red circles ($r \in [6.35, 10.42]$ px) are foreground sunflower seed heads. False detections of trees near horizon are avoided.

2. Line and Circle Fitting using RANSAC

In this task, we aim to fit a line and ultimately a circle to a set of 100 noisy points using RANSAC. First, 50 points conforming to a circle ($r = 10$, center $(2, 3)$) and 50 points conforming to a line ($y = -x + 2$) are generated as given in Listing 1 in the assignment document.

[22]:
```
import numpy as np
import matplotlib.pyplot as plt
```

```
np.random.seed(0)
N = 100
half_n = N // 2

# Circle points with radial noise
r_gt, x0_gt, y0_gt = 10.0, 2.0, 3.0
s_circ = r_gt / 16.0  # std = 0.625
t = np.random.uniform(0, 2 * np.pi, half_n)
n_circ = s_circ * np.random.randn(half_n)
x_c = x0_gt + (r_gt + n_circ) * np.cos(t)
y_c = y0_gt + (r_gt + n_circ) * np.sin(t)
X_circ = np.hstack((x_c.reshape(half_n, 1), y_c.reshape(half_n, 1)))

# Line points with vertical noise
s_line = 1.0  # std = 1.0
m_gt, b_gt = -1.0, 2.0
x_l = np.linspace(-12, 12, half_n)
y_l = m_gt * x_l + b_gt + s_line * np.random.randn(half_n)
X_line = np.hstack((x_l.reshape(half_n, 1), y_l.reshape(half_n, 1)))

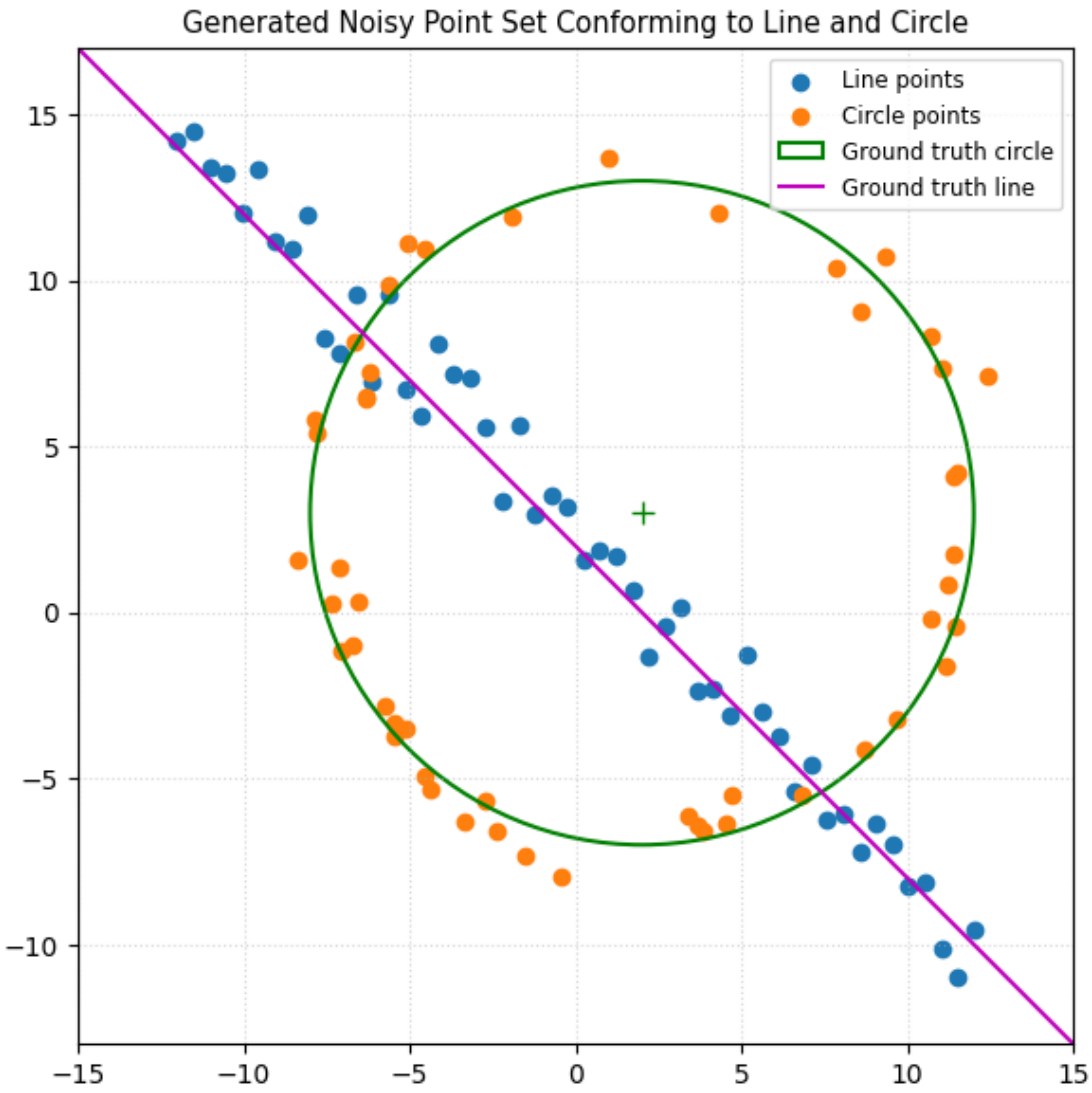
X = np.vstack((X_circ, X_line))

# Plot generated points with ground truth
fig, ax = plt.subplots(figsize=(6.0, 6.0))
ax.scatter(X_line[:, 0], X_line[:, 1], color='#1f77b4', label='Line points')
ax.scatter(X_circ[:, 0], X_circ[:, 1], color='#ff7f0e', label='Circle points')

circle_gt = plt.Circle((x0_gt, y0_gt), r_gt, color='g', fill=False, linewidth=1.5, label='Ground␣
 ↪truth circle')
ax.add_patch(circle_gt)
ax.plot(x0_gt, y0_gt, '+', color='g', markersize=8)

x_vals = np.array([-15, 15])
ax.plot(x_vals, m_gt * x_vals + b_gt, color='m', linewidth=1.5, label='Ground truth line')

ax.set_xlim(-15, 15); ax.set_ylim(-13, 17); ax.set_aspect('equal')
ax.grid(True, linestyle=':', alpha=0.5)
ax.legend(loc='upper right', fontsize=8.5)
ax.set_title("Generated Noisy Point Set Conforming to Line and Circle", fontsize=10.5, pad=6)
plt.tight_layout()
plt.show()
```



2.1 Threshold Selection and Fitting Procedure

- **Line Fitting:** A line is represented as $ax + by - d = 0$ with constraint $\|[a, b]^\top\|_2 = 1$. The orthogonal distance threshold is set to $\tau_{\text{line}} = 1.5$. Consensus threshold is set to $T_{\text{line}} = 35$ points. The consensus inliers are then refined using Total Least Squares (TLS) via single value decomposition (SVD).
- **Circle Fitting:** The inliers of the best line are subtracted. The radial distance threshold is set to $\tau_{\text{circ}} = 1.50$ $(2.4\sigma_{\text{circ}})$ with consensus threshold $T_{\text{circ}} = 30$ points. The consensus inliers are refined using Nelder Mead optimization (`scipy.optimize.minimize`) on radial loss.

[23]:
```
from scipy.optimize import minimize

# RANSAC Line Estimation
def fit_line_2pts(p1, p2):
```

```
    dx, dy = p2[0] - p1[0], p2[1] - p1[1]
    norm = np.hypot(dx, dy)
    return np.array([-dy / norm, dx / norm, (-dy * p1[0] + dx * p1[1]) / norm]) if norm > 1e-8␣
 ↪else None

np.random.seed(42)
best_line_inl, sample_line_idx, sample_line_model = [], None, None
for _ in range(1000):
    idx = np.random.choice(len(X), 2, replace=False)
    m = fit_line_2pts(X[idx[0]], X[idx[1]])
    if m is None: continue
    inl = np.where(np.abs(m[0] * X[:, 0] + m[1] * X[:, 1] - m[2]) < 1.5)[0]
    if len(inl) > len(best_line_inl):
        best_line_inl, sample_line_idx, sample_line_model = inl, idx, m

# Optimize the line using Total Least Squares
p_mean = np.mean(X[best_line_inl], axis=0)
normal_tls = np.linalg.svd(X[best_line_inl] - p_mean)[2][-1]
best_fit_line = np.array([normal_tls[0], normal_tls[1], np.dot(normal_tls, p_mean)])

# Subtract line consensus and fit circle on remaining points
rem_idx = np.where(~np.isin(np.arange(len(X)), best_line_inl))[0]
X_rem = X[rem_idx]

def fit_circle_3pts(p1, p2, p3):
    A = np.array([[2*(p2[0]-p1[0]), 2*(p2[1]-p1[1])], [2*(p3[0]-p2[0]), 2*(p3[1]-p2[1])]])
    B = np.array([p2[0]**2 + p2[1]**2 - (p1[0]**2 + p1[1]**2), p3[0]**2 + p3[1]**2 - (p2[0]**2 +␣
 ↪p2[1]**2)])
    if abs(np.linalg.det(A)) < 1e-6: return None
    xc, yc = np.linalg.solve(A, B)
    return np.array([xc, yc, np.hypot(p1[0]-xc, p1[1]-yc)])

best_circ_rel, sample_circ_rel_idx, sample_circ_model = [], None, None
for _ in range(2000):
    idx = np.random.choice(len(X_rem), 3, replace=False)
    m = fit_circle_3pts(X_rem[idx[0]], X_rem[idx[1]], X_rem[idx[2]])
    if m is None or m[2] > 50.0: continue
    inl = np.where(np.abs(np.hypot(X_rem[:, 0] - m[0], X_rem[:, 1] - m[1]) - m[2]) < 1.50)[0]
    if len(inl) > len(best_circ_rel):
        best_circ_rel, sample_circ_rel_idx, sample_circ_model = inl, idx, m

# Optimize the circle via Nelder Mead optimization
def loss(p): return np.sum((np.hypot(X_rem[best_circ_rel, 0] - p[0], X_rem[best_circ_rel, 1] -␣
 ↪p[1]) - p[2]) ** 2)
best_fit_circle = minimize(loss, sample_circ_model, method='Nelder-Mead').x
circ_inl = rem_idx[best_circ_rel]
sample_circ_idx = rem_idx[sample_circ_rel_idx]

# Comparison Table
print(f"{'Model':<8}{'Parameter':<18}{'Ground Truth':<16}{'Sample':<16}{'Refined Fit':<14}")
print("-" * 72)
print(f"{'Line':<8}{'Normal [a, b]':<18}{'[0.707, 0.707]':<16}{f'[{sample_line_model[0]:.3f},␣
 ↪{sample_line_model[1]:.3f}]':<16}{f'[{best_fit_line[0]:.3f}, {best_fit_line[1]:.3f}]':<14}")
print(f"{'':<8}{'Distance d':<18}{'1.414':<16}{f'{sample_line_model[2]:.3f}':
 ↪<16}{f'{best_fit_line[2]:.3f}':<14}")
print(f"{'':<8}{'Inliers':<18}{'50':<16}{'--':<16}{f'{len(best_line_inl)}':<14}")
print("-" * 72)
```

```
print(f"{'Circle':<8}{'Center (x, y)':<18}{'(2.00, 3.00)':<16}{f'({sample_circ_model[0]:.2f},␣
 ↪{sample_circ_model[1]:.2f})':<16}{f'({best_fit_circle[0]:.2f}, {best_fit_circle[1]:.2f})':
 ↪<14}")
print(f"{'':<8}{'Radius r':<18}{'10.00':<16}{f'{sample_circ_model[2]:.2f}':
 ↪<16}{f'{best_fit_circle[2]:.2f}':<14}")
print(f"{'':<8}{'Inliers':<18}{'50':<16}{'--':<16}{f'{len(circ_inl)}':<14}")

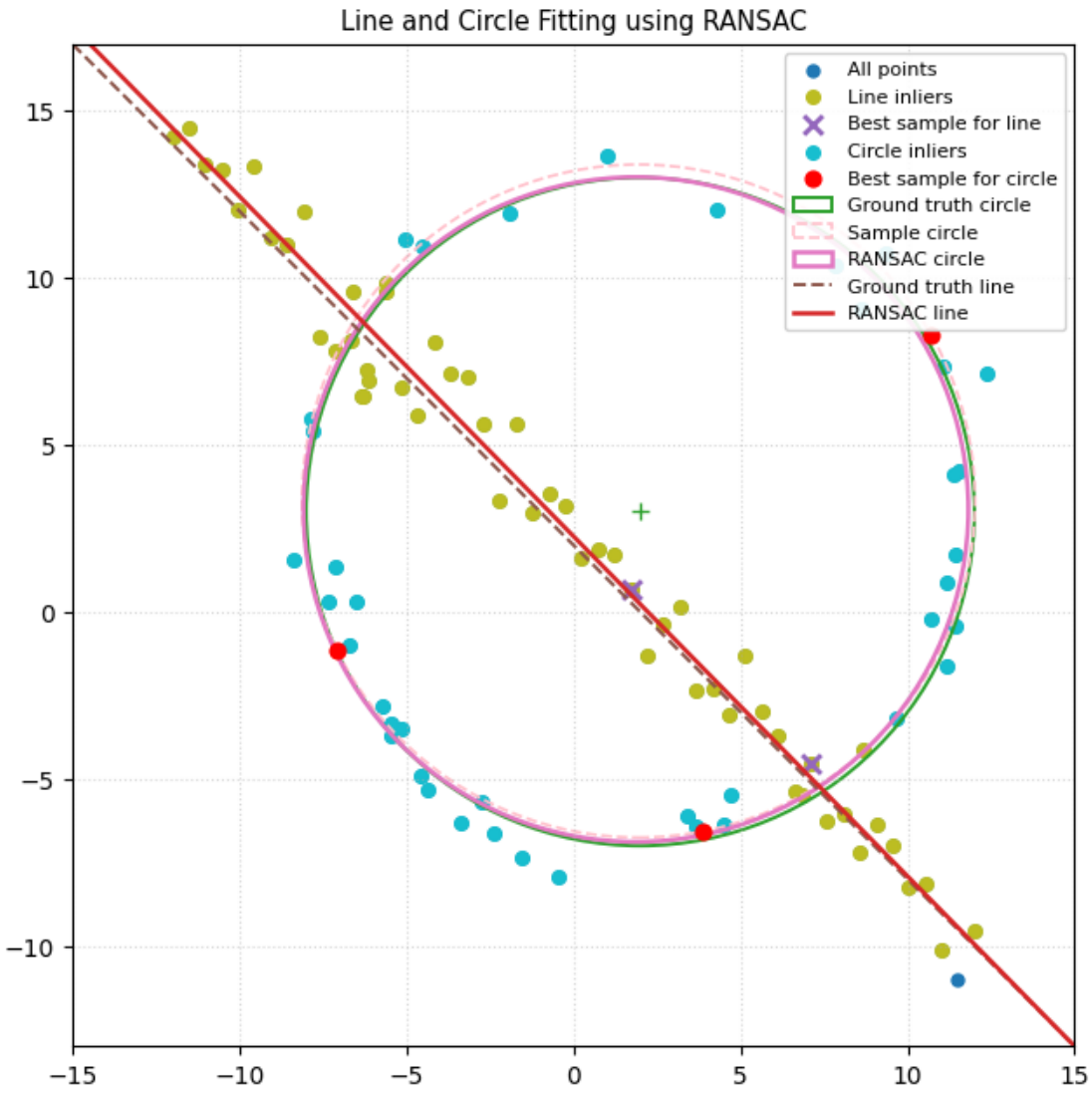
# Plot results
fig, ax = plt.subplots(figsize=(6.5, 6.5))
ax.scatter(X[:, 0], X[:, 1], color='#1f77b4', s=25, label='All points')
ax.scatter(X[best_line_inl, 0], X[best_line_inl, 1], color='#bcbd22', s=30, label='Line inliers')
ax.scatter(X[sample_line_idx, 0], X[sample_line_idx, 1], color='#9467bd', marker='x', s=60,␣
 ↪linewidths=2, label='Best sample for line')
ax.scatter(X[circ_inl, 0], X[circ_inl, 1], color='#17becf', s=30, label='Circle inliers')
ax.scatter(X[sample_circ_idx, 0], X[sample_circ_idx, 1], color='red', s=40, zorder=5,␣
 ↪label='Best sample for circle')

ax.add_patch(plt.Circle((x0_gt, y0_gt), r_gt, color='#2ca02c', fill=False, linewidth=1.4,␣
 ↪label='Ground truth circle'))
ax.plot(x0_gt, y0_gt, '+', color='#2ca02c', markersize=7)
ax.add_patch(plt.Circle((sample_circ_model[0], sample_circ_model[1]), sample_circ_model[2],␣
 ↪color='pink', linestyle='--', fill=False, linewidth=1.2, label='Sample circle'))
ax.add_patch(plt.Circle((best_fit_circle[0], best_fit_circle[1]), best_fit_circle[2],␣
 ↪color='#e377c2', fill=False, linewidth=1.8, label='RANSAC circle'))

xr = np.array([-15, 15])
ax.plot(xr, m_gt * xr + b_gt, color='#8c564b', linestyle='--', linewidth=1.4, label='Ground␣
 ↪truth line')
ax.plot(xr, (best_fit_line[2] - best_fit_line[0] * xr) / best_fit_line[1], color='#d62728',␣
 ↪linewidth=1.8, label='RANSAC line')

ax.set_xlim(-15, 15); ax.set_ylim(-13, 17); ax.set_aspect('equal')
ax.grid(True, linestyle=':', alpha=0.5)
ax.legend(loc='upper right', fontsize=8, framealpha=0.9)
ax.set_title("Line and Circle Fitting using RANSAC", fontsize=10.5, pad=6)
plt.tight_layout()
plt.show()
```

```
Model   Parameter         Ground Truth    Sample          Refined Fit
-------------------------------------------------------------------------
Line    Normal [a, b]     [0.707, 0.707]  [-0.696, -0.718][0.713, 0.701]
        Distance d        1.414           -1.669          1.588
        Inliers           50              --              56
-------------------------------------------------------------------------
Circle  Center (x, y)     (2.00, 3.00)    (1.95, 3.32)    (1.87, 3.07)
        Radius r          10.00           10.07           9.95
        Inliers           50              --              43
```



2.2 (d) What happens if we fit the circle first?

If we fit the circle first, the line cuts through the circle like a secant. The line points that lie near the circle boundary fall within the circle error threshold, so the circle RANSAC falsely grabs some line points as circle inliers, which biases the circle center and radius. Moreover, subtracting the circle consensus removes the middle section of the line, leaving it broken into two separate parts which hinders the subsequent line fitting. In contrast, fitting the line first is safe because a line can intersect a circle at at most 2 points, so it cannot break the circle.

3. Planar Homography and Image Warping

3.1 Methodology

A planar homography $\mathbf{x}' \sim \mathbf{H}\mathbf{x}$ defines a projective mapping with 8 degrees of free-

dom. Four coplanar point correspondences determine $\mathbf{H}$ via Direct Linear Transformation (`cv2.getPerspectiveTransform`). Inverse backward mapping with bilinear interpolation (`cv2.warpPerspective`) is used to warp the overlay without holes. Additive blending ($\mathbf{I}_{\text{blend}} = \mathbf{I}_{\text{target}} + 0.5\,\mathbf{I}_{\text{warped}}$) replicates Figure 2, preserving underlying stonework and architectural texture.

[24]:
```
# Task 3: Planar Homography Warping on Architectural Facades
def superimpose_image(target_path, overlay_path, dst_pts, blend_mode='additive', alpha=0.5):
    target = cv2.imread(target_path)
    overlay = cv2.imread(overlay_path, cv2.IMREAD_UNCHANGED)
    h_ov, w_ov = overlay.shape[:2]
    src_pts = np.float32([[0, 0], [w_ov - 1, 0], [w_ov - 1, h_ov - 1], [0, h_ov - 1]])

    H = cv2.getPerspectiveTransform(src_pts, dst_pts.astype(np.float32))
    h_t, w_t = target.shape[:2]
    warped = cv2.warpPerspective(overlay, H, (w_t, h_t))

    mask = np.zeros((h_t, w_t), dtype=np.uint8)
    cv2.fillConvexPoly(mask, dst_pts.astype(np.int32), 255)

    result = target.copy()
    if blend_mode == 'additive':
        result[mask > 0] = cv2.add(result[mask > 0], (warped[mask > 0, :3] * alpha).astype(np.
 ↪uint8))
    return result, H

# Pair 1: Wadham College Left Facade + UK Flag (Figure 2 Replication)
pts_wadham = np.float32([[147, 208], [517, 292], [521, 521], [135, 519]])
res_wadham, H1 = superimpose_image('src/wadham.jpg', 'src/flag.png', pts_wadham)

orig_wadham = cv2.imread('src/wadham.jpg')
for i, pt in enumerate(pts_wadham):
    cv2.circle(orig_wadham, tuple(pt.astype(int)), 6, (0, 0, 255), -1)
    cv2.putText(orig_wadham, f"P{i+1}", (int(pt[0]) + 8, int(pt[1]) + 5), cv2.
 ↪FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 255), 2)
cv2.polylines(orig_wadham, [pts_wadham.astype(np.int32)], True, (0, 255, 255), 2)

# Pair 2: Wadham College Right Facade + Sri Lanka Flag
pts_sl = np.float32([[521, 292], [980, 190], [980, 582], [521, 521]])
res_sl, H2 = superimpose_image('src/wadham.jpg', 'src/sri_lanka_flag.png', pts_sl)

orig_sl = cv2.imread('src/wadham.jpg')
for i, pt in enumerate(pts_sl):
    cv2.circle(orig_sl, tuple(pt.astype(int)), 6, (0, 0, 255), -1)
    cv2.putText(orig_sl, f"P{i+1}", (int(pt[0]) + 8, int(pt[1]) + 5), cv2.FONT_HERSHEY_SIMPLEX,␣
 ↪0.6, (0, 255, 255), 2)
cv2.polylines(orig_sl, [pts_sl.astype(np.int32)], True, (0, 255, 255), 2)

print("Pair 1 Homography Matrix (UK Flag -> Wadham Left Facade):")
print(np.array2string(H1, precision=4, suppress_small=True))
print("\nPair 2 Homography Matrix (Sri Lanka Flag -> Wadham Right Facade):")
print(np.array2string(H2, precision=4, suppress_small=True))

fig, axes = plt.subplots(2, 2, figsize=(11, 8.5))
axes[0, 0].imshow(cv2.cvtColor(orig_wadham, cv2.COLOR_BGR2RGB)); axes[0, 0].set_title("Pair 1:␣
 ↪Wadham Left Facade (4 Corners)", fontsize=10); axes[0, 0].axis('off')
axes[0, 1].imshow(cv2.cvtColor(res_wadham, cv2.COLOR_BGR2RGB)); axes[0, 1].set_title("Pair 1:␣
 ↪Superimposed UK Flag (Fig. 2 Replication)", fontsize=10); axes[0, 1].axis('off')
```

```
axes[1, 0].imshow(cv2.cvtColor(orig_sl, cv2.COLOR_BGR2RGB)); axes[1, 0].set_title("Pair 2:␣
 ↪Wadham Right Facade (4 Corners)", fontsize=10); axes[1, 0].axis('off')
axes[1, 1].imshow(cv2.cvtColor(res_sl, cv2.COLOR_BGR2RGB)); axes[1, 1].set_title("Pair 2:␣
 ↪Superimposed Sri Lanka Flag", fontsize=10); axes[1, 1].axis('off')
plt.tight_layout()
plt.show()
```

```
Pair 1 Homography Matrix (UK Flag -> Wadham Left Facade):
[[  0.2894  -0.0189 147.    ]
 [  0.0983   0.2999 208.    ]
 [  0.0002  -0.       1.    ]]

Pair 2 Homography Matrix (Sri Lanka Flag -> Wadham Right Facade):
[[  0.0403   0.     521.    ]
 [ -0.1415   0.3584 292.    ]
 [ -0.0003   0.       1.    ]]
```



3.2 Non-Technical Rationale and Observations

- **Pair 1 (Left Facade & UK Flag):** Selected corners $P_1(147, 208)$, $P_2(517, 292)$, $P_3(521, 521)$, $P_4(135, 519)$ on the residential wing. Non-technical rationale: celebrating English collegiate heritage at Wadham College, Oxford. Additive blending retains stone mullions and masonry shadows.

- **Pair 2 (Right Facade & Sri Lanka Flag):** Selected corners $P_1(521, 292)$, $P_2(980, 190)$, $P_3(980, 582)$, $P_4(521, 521)$ on the perpendicular chapel wing. Non-technical rationale: academic diplomacy celebrating Sri Lankan scholars studying at Oxford. Captures the opposite perspective vanishing direction (receding to the right) while retaining Gothic traceried windows.

4. PCB Alignment and Defect Detection

4.1 Methodology

In this task, we align a set of PCB images to a reference template (`AECC1.jpg`) and compute difference images. SIFT keypoints are detected in both images using `cv2.SIFT_create()`, and descriptors are matched with `cv2.BFMatcher()`. Good matches are filtered using Lowe's ratio test ($d_1/d_2 < 0.75$), and a planar homography is estimated using RANSAC (`cv2.findHomography`). The target image is warped onto the template coordinate frame using `cv2.warpPerspective`, and absolute difference is computed via `cv2.absdiff`.

[25]:
```python
# Task 4: SIFT Feature Matching, Homography Alignment, and Difference Inspection
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt

pcb_dir = 'src/pcbs'
template_bgr = cv2.imread(os.path.join(pcb_dir, 'AECC1.jpg'))
template_gray = cv2.cvtColor(template_bgr, cv2.COLOR_BGR2GRAY)
h_t, w_t = template_gray.shape

sift = cv2.SIFT_create()
kp_t, des_t = sift.detectAndCompute(template_gray, None)

targets = ['AAAA1.jpg', 'ADAD1.jpg', 'AECC2.jpg', 'AECC3.jpg', 'AECC4.jpg']
results = {}

for name in targets:
    target_bgr = cv2.imread(os.path.join(pcb_dir, name))
    target_gray = cv2.cvtColor(target_bgr, cv2.COLOR_BGR2GRAY)

    kp_tgt, des_tgt = sift.detectAndCompute(target_gray, None)

    bf = cv2.BFMatcher()
    matches = bf.knnMatch(des_tgt, des_t, k=2)
    good = [m for m, n in matches if m.distance < 0.75 * n.distance]

    src_pts = np.float32([kp_tgt[m.queryIdx].pt for m in good]).reshape(-1, 1, 2)
    dst_pts = np.float32([kp_t[m.trainIdx].pt for m in good]).reshape(-1, 1, 2)

    H, mask = cv2.findHomography(src_pts, dst_pts, cv2.RANSAC, 5.0)
    inliers = [good[idx] for idx, val in enumerate(mask.ravel()) if val == 1]

    warped_bgr = cv2.warpPerspective(target_bgr, H, (w_t, h_t))
    diff_bgr = cv2.absdiff(template_bgr, warped_bgr)
    diff_gray = cv2.cvtColor(diff_bgr, cv2.COLOR_BGR2GRAY)

    mask_ov = cv2.warpPerspective(np.ones_like(target_gray) * 255, H, (w_t, h_t)) > 200
```

```
    mean_val = float(np.mean(diff_gray[mask_ov]))

    results[name] = {
        'kp_target': kp_tgt,
        'good': good,
        'inliers': inliers,
        'diff_gray': diff_gray,
        'mask_ov': mask_ov,
        'mean_diff': mean_val
    }

print(f"{'Image':<12}{'Keypoints':<12}{'Good Matches':<14}{'RANSAC Inliers':<16}{'Inlier %':
 ↪<12}{'Mean Diff':<10}")
print("-" * 76)
for name in targets:
    r = results[name]
    pct = (len(r['inliers']) / len(r['good'])) * 100 if r['good'] else 0
    print(f"{name:<12}{len(r['kp_target']):<12}{len(r['good']):<14}{len(r['inliers']):<16}{pct:
 ↪<12.1f}{r['mean_diff']:<10.2f}")

# Display feature matches (left) and difference images (right) for all 5 PCB images
fig, axes = plt.subplots(len(targets), 2, figsize=(9.5, 11.5))

for i, name in enumerate(targets):
    r = results[name]
    target_bgr = cv2.imread(os.path.join(pcb_dir, name))

    match_vis = cv2.drawMatches(target_bgr, r['kp_target'], template_bgr, kp_t,
                                r['inliers'][:30], None, flags=cv2.
 ↪DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS)

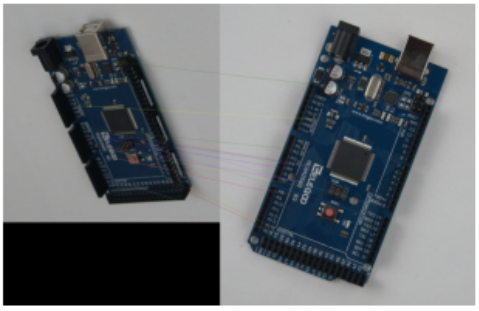
    axes[i, 0].imshow(cv2.cvtColor(match_vis, cv2.COLOR_BGR2RGB))
    axes[i, 0].set_title(f"{name} Matches ({len(r['inliers'])} inliers / {len(r['good'])}␣
 ↪good)", fontsize=10, pad=4)
    axes[i, 0].axis('off')

    disp_diff = r['diff_gray'].copy()
    disp_diff[~r['mask_ov']] = 0
    im = axes[i, 1].imshow(disp_diff, cmap='inferno', vmin=0, vmax=120)
    axes[i, 1].set_title(f"{name} Difference Image (Mean: {r['mean_diff']:.1f})", fontsize=10,␣
 ↪pad=4)
    axes[i, 1].axis('off')

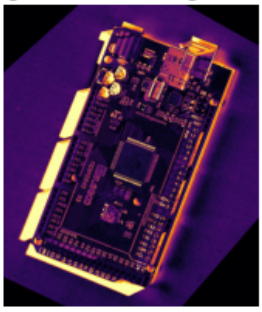
plt.tight_layout()
plt.show()
```

```
Image       Keypoints   Good Matches  RANSAC Inliers  Inlier %    Mean Diff
----------------------------------------------------------------------------
AAAA1.jpg   2414        51            17              33.3        28.74
ADAD1.jpg   8486        369           194             52.6        31.23
AECC2.jpg   12315       4322          3844            88.9        3.44
AECC3.jpg   15198       5629          5001            88.8        3.19
AECC4.jpg   11610       3830          3393            88.6        3.94
```

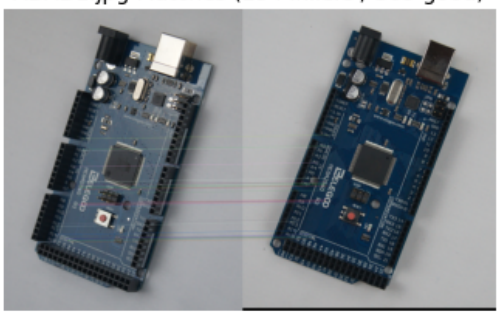
AAAA1.jpg Matches (17 inliers / 51 good)



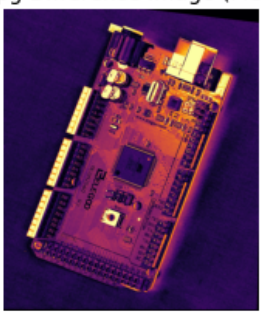
AAAA1.jpg Difference Image (Mean: 28.7)



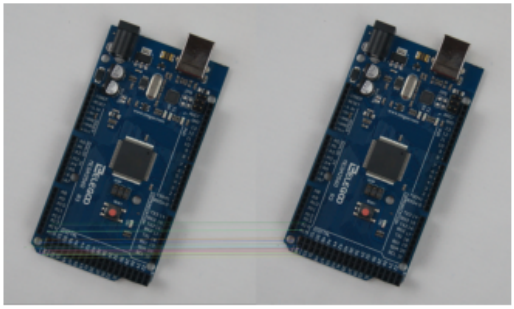
ADAD1.jpg Matches (194 inliers / 369 good)



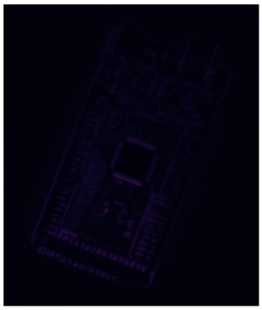
ADAD1.jpg Difference Image (Mean: 31.2)



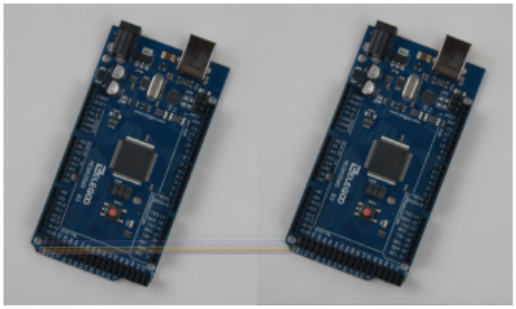
AECC2.jpg Matches (3844 inliers / 4322 good)



AECC2.jpg Difference Image (Mean: 3.4)



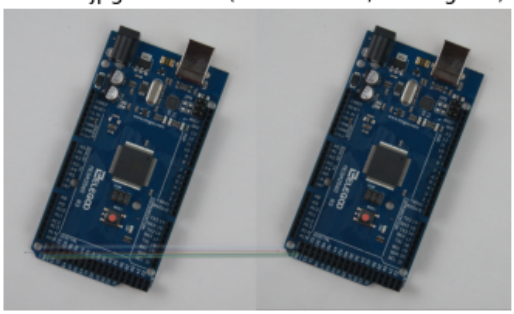
AECC3.jpg Matches (5001 inliers / 5629 good)



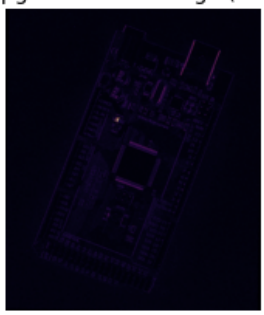
AECC3.jpg Difference Image (Mean: 3.2)



AECC4.jpg Matches (3393 inliers / 3830 good)



AECC4.jpg Difference Image (Mean: 3.9)



4.2 Observations and Analysis of High Difference Elements

- **High Differences in 3D Components (e):** Across images (most noticeably `ADAD1` and `AAAA1`), the USB Type-B port, electrolytic capacitors, and reset button show distinct bright difference regions. This occurs because planar homography assumes all scene features lie on a single 2D plane ($Z = 0$, the flat board surface). These components extend a few millimeters above the board in 3D space ($Z > 0$). When the viewpoint or board orientation changes, 3D parallax causes the tops and sides of these tall parts to project to different positions than what a 2D planar transform predicts. Furthermore, changes in angle alter cast shadows and specular reflections on the metal USB casing.
- **Board Variations:** Boards `AECC2`, `AECC3`, and `AECC4` are identical models photographed in nearly matching alignments, resulting in very low mean differences (3.2–3.9). `AAAA1` and `ADAD1` have different orientations and board layouts, leading to higher differences (28.7 and 31.2).